doi: 10.2965/jwet.22-058 Journal of Water and Environment Technology, Vol.20, No.6: 219–237, 2022

Original Article

Development and Validation of an Analytical Method for Simultaneous Determination of Perfluoroalkyl Acids in Drinking Water by Liquid Chromatography/Tandem Mass Spectrometry

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ABSTRACT

The environmental presence and drinking water contamination of per- and polyfluoroalkyl substances (PFAS) have been reported since the early 2000s. This study seeks to develop a liquid chromatography/tandem mass spectrometry analytical method for the simultaneous determination of 21 perfluoroalkyl acids (PFAAs) in drinking water to support future regulations in Japan. Inter-laboratory tests were conducted in 16 laboratories using different instrument to verify the applicability of the developed method for a wide range of drinking water samples in Japan. Recovery tests of PFAA-fortified tap water samples obtained in each laboratory were conducted at set points of 1 and 10 ng/L. Calibration curve linearity, trueness (recovery), repeatability (RSD_r), and reproducibility (RSD_R) at these analyte concentrations were calculated using data obtained from the recovery tests. The trueness, RSD_r, and RSD_R of most PFAA analytes were satisfactory when the recoveries were corrected by ^{13}C-PFAA extraction standards with similar recovery to the corresponding PFAS analytes. The developed analytical method is valid for the quantification of the target PFAAs in drinking water. However, satisfactory PFAA quantification requires recovery adjustment using surrogates with similar recovery characteristics to the PFAA analytes.

Keywords: per- and polyfluoroalkyl substances (PFAS), perfluoroalkyl acids (PFAAs), drinking water, LC/MS/MS, validation

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Received: May 26, 2022, Accepted: September 29, 2022, Published online: December 10, 2022

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INTRODUCTION

The environmental presence and drinking water contamination of per- and polyfluoroalkyl substances (PFAS) have been widely reported since the early 2000s [1–4]. PFAS in drinking water have become a serious global concern, because of their widespread environmental occurrence, high level of persistence and bioaccumulation, and toxic properties. In 2016, the United States Environmental Protection Agency (U.S. EPA) established a non-regulatory drinking water health advisory of 70 ng/L for individual and total concentrations of perfluorooctanesulfonic acid (PFOS) and perfluorooctanoic acid (PFOA). These 8-carbon perfluoroalkyl acids (PFAAs) are the most extensively investigated PFAS. The Stockholm Convention on persistent organic pollutants (POPs) [5] also has banned or restricted the production and the use of PFOS and PFOA. Further, the Stockholm Convention on POPs recommended to add perfluorohexanesulfonic acid (PFHxS) to this category in 2019.

In Japan, PFOS and PFOA have been assigned to the third-priority category in drinking water quality control as "items for further study" [6]. However, these two PFAAs were upgraded to the second-priority category as "complementary items" in April of 2020 according to the Stockholm Convention on POPs. A tentative target value of 50 ng/L was established for the total concentration of PFOS and PFOA at that time [7]. PFHxS was added to the "items for further study" category in April 2021, although a target value has yet to be established.

As the restrictions on PFOS, PFOA, and PFHxS have become more severe, alternative PFAS have been widely explored and used. The contamination of drinking water by such alternative PFAS in the U.S. and other countries has recently been reported [8–11]. In response, the European Union (EU) restricted the use of PFAS containing a perfluoroalkyl moiety with three or more carbons (*i.e.*, $-C_nF_{2n}-$, $n \geq 3$) or a perfluoroalkylether moiety with two or more carbons (*i.e.*, $-C_nF_{2n}OC_mF_{2m}-$, n and $m \geq 1$) in 2020. Twenty PFAS, perfluoroalkyl sulfonic acids (PFSAs) and perfluoroalkyl carboxylic acids (PFCAs) with carbon chain lengths of 4–13, were selected as analytical targets by the EU [12]. Ten states in the U.S., 10 have developed guidelines for other PFAS (primarily PFAAs), and additional state guidelines are expected [13].

PFAS are hydrophobic substances, which raises concerns regarding human exposure to PFAS via drinking water. However, information on drinking water contamination and human exposure to PFAS other than PFOS and PFOA is limited in Japan. This is mainly due to the lack of Japanese standards for analytical methods to determine PFAS other than PFOS, PFOA [14], and PFHxS [15] in drinking water. More recently, standard analytical methods for other PFAS were published by the International Organization for Standardization (ISO) and U.S. EPA [16–19]: however, the applicability of these analytical methods for the determination of PFAAs in drinking water in Japan has not been evaluated or reported.

To overcome this, a liquid chromatography/tandem mass spectrometry (LC/MS/MS) analytical method is developed for the simultaneous determination of 21 PFAAs with the goal of supporting future drinking water regulations in Japan. The applicability of the analytical method for PFAA determination in drinking water in Japan is evaluated by conducting an inter-laboratory test performed in 16 laboratories by using different instruments.

MATERIALS AND METHODS

Analytical targets

Twenty-one PFAAs comprising 8 PFSAs with carbon-atom chain lengths of 4–10 and 12 and 13 PFCAs with carbon-atom chain lengths of 4–14, 16, and 18, were selected as analytical targets (**Table 1**). Although these PFAAs have linear and branched isomers, only the linear isomer of each PFAA was assigned because it is the main component of standard mixtures used in the study.

Development of LC/MS/MS analytical conditions

A standard solution of each PFAA was measured by LC/MS/MS (LCMS-8050, Shimadzu Corporation,Kyoto, Japan) via the electrospray ionization (ESI) in the total-ion monitoring (TIM) mode to acquire the mass spectrum of each PFAA. The ion exhibiting the highest intensity was chosen as the precursor ion. Product ions generated from the precursor ion with high intensities were chosen as quantifier and qualifier ions.

After identifying the quantifier and qualifier ions of each PFAA, the mobile-phase gradient conditions for LC were optimized using an octadecyl silica (ODS) column (Inert-Sustain AQ-C18, 2.1 x 150 mm, pore size 3 μm, GL Sciences Inc., Tokyo, Japan), and a mobile-phase gradient comprising 10 mM aqueous ammonium acetate solution and acetonitrile.

Development of the pretreatment method

The pretreatment method for the PFAA analytes (**Fig. 1**) was developed based on existing standard analytical

Table 1 Analytical targets.

#	Category	Analyte name	Abbreviation	CAS RN	Molecular formula	Structural formula	Molecular weight
1	Perfluoroalkyl Sulfonates (PFSAs)	Perfluorobutanesulfonic acid	PFBS	375–73-5	$C_4HF_9SO_3$	$F_3C(CF_2)_3SO_3H$	300.1
2		Perfluoropentanesulfonic acid	PFPeS	2706–91-4	$C_5HF_{11}SO_3$	$F_3C(CF_2)_4SO_3H$	350.11
3		Perfluorohexanesulfonic acid	PFHxS	355–46-4	$C_6HF_{13}SO_3$	$F_3C(CF_2)_5SO_3H$	400.11
4		Perfluoroheptanesulfonic acid	PFHpS	375–92-8	$C_7HF_{15}SO_3$	$F_3C(CF_2)_6SO_3H$	450.12
5		Perfluorooctanesulfonic acid	PFOS	1763–23-1	$C_8HF_{17}SO_3$	$F_3C(CF_2)_7SO_3H$	500.13
6		Perfluorononanesulfonic acid	PFNS	68259–12-1	$C_9HF_{19}SO_3$	$F_3C(CF_2)_8SO_3H$	550.14
7		Perfluorodecanesulfonic acid	PFDS	335–77-3	$C_{10}HF_{21}SO_3$	$F_3C(CF_2)_9SO_3H$	600.14
8		Perfluorododecaneculfonic acid	PFDoDS	79780–39-5	$C_{12}HF_{25}SO_3$	$F_3C(CF_2)_{11}SO_3H$	700.16
9	Perfluoroalkyl carboxylic acids (PFCAs)	Perfluorobutanoic acid	PFBA	375–22-4	$C_4HF_7O_2$	$F_3C(CF_2)_2COOH$	214.04
10		Perfluoropentanoic acid	PFPeA	2706–90-3	$C_5HF_9O_2$	$F_3C(CF_2)_3COOH$	264.05
11		Perfluorohexanoic acid	PFHxA	307–24-4	$C_6HF_{11}O_2$	$F_3C(CF_2)_4COOH$	314.05
12		Perfluoroheptanoic acid	PFHpA	375–85-9	$C_7HF_{13}O_2$	$F_3C(CF_2)_5COOH$	364.06
13		Perfluorooctanoic acid	PFOA	335–67-1	$C_8HF_{15}O_2$	$F_3C(CF_2)_6COOH$	414.07
14		Perfluorononanoic acid	PFNA	375–95-1	$C_9HF_{17}O_2$	$F_3C(CF_2)_7COOH$	464.08
15		Perfluorodecanoic acid	PFDA	335–76-2	$C_{10}HF_{19}O_2$	$F_3C(CF_2)_8COOH$	514.08
16		Perfluoroundecanoic acid	PFUnDA	2058–94-8	$C_{11}HF_{21}O_2$	$F_3C(CF_2)_9COOH$	564.09
17		Perfluorododecanoic acid	PFDoDA	307–55-1	$C_{12}HF_{23}O_2$	$F_3C(CF_2)_{10}COOH$	614.1
18		Perfluorotridecanoic acid	PFTrDA	72629–94-8	$C_{13}HF_{25}O_2$	$F_3C(CF_2)_{11}COOH$	664.11
19		Perfluorotetradecanoic acid	PFTeDA	376–06-7	$C_{14}HF_{27}O_2$	$F_3C(CF_2)_{12}COOH$	714.11
20		Perfluorohexadecanoic acid	PFHxDA	67905–19-5	$C_{16}HF_{31}O_2$	$F_3C(CF_2)_{14}COOH$	814.13
21		Perfluorooctadecanoic acid	PFOcDA	16517–11-6	$C_{18}HF_{35}O_2$	$F_3C(CF_2)_{16}COOH$	914.14

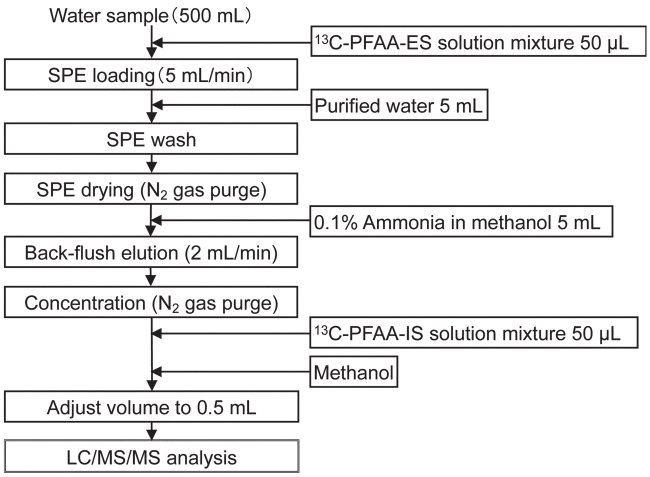


Fig. 1 Flowchart of the pretreatment method.

procedures such as those of the ISO and U.S. EPA [16–19]. A 500 mL water sample was concentrated and purified by solid-phase extraction (SPE) to 0.5 mL (1,000x concentration), after adding 13 ^{13}C-PFAA extraction standards (ES) as surrogates. The prepared samples were analyzed by LC/MS/MS. Losses sustained during pretreatment were corrected by the ^{13}C-PFAA-ES. Four additional ^{13}C-PFAA injection standards (IS) were added after the SPE, to calculate the recovery of each PFAA analyte and the ^{13}C-PFAA-ES from the pretreatment process.

When this validation test was conducted in 2000, target values for PFOS and PFOA had not been established in Japan. However, it is essential to conduct the test for lower PFOS and PFOA concentrations than those established by the criterion in foreign countries. Thus, the objective of this study is to develop an analytical method with a 1 ng/L limit of quantification (LOQ) for each PFAA in a water sample.

Method validation

An inter-laboratory test was conducted using the standard operation procedure (SOP) developed in 16 laboratories to confirm the applicability of the pretreatment and LC/MS/MS analytical methods toward the analysis of various drinking water samples and instrumentation. The number of laboratories participated in inter-laboratory tests required to identify a developed methodology as a standard method for testing drinking water has not been established in Japan. However, the validation of an analytical method by inter-laboratory testing should be involve participation by threshold number of laboratories to achieve international recognition. For example, the Codex standard requires 8 or more valid analytical results for quantitative analysis [20,21]. In this study, 15 laboratories in addition to the National Institute of Health Sciences (NIHS) participated in the recovery tests using tap water. These organizations included 4 laboratories from the prefectural institutes of public health (Osaka Institute of Public Health, Tokyo Metropolitan Institute of Public Health, Saitama Institute of Public Health, and Kanagawa Prefectural Institute of Public Health), 5 laboratories from the waterworks bureaus (The City of Hiroshima Waterworks Bureau, City of Kawasaki Waterworks Bureau, Fukuyama City Waterworks and Sewerage Bureau, Kanagawa Prefectural Government, and Nagoya City Waterworks and Sewerage Bureau), 3 laboratories from the registered analytical laboratories (Mie Prefecture Environmental Conservation Agency, Chiba Pharmaceutical Association, and Gifu Research Center for Public Health), and 3 laboratories from the analytical instrumentation manufacturers (Nihon Waters K.K., GL Sciences Inc., and Agilent Technologies Japan, Ltd.).

The sample pretreatment method, concentration ranges of the calibration curves, and water sample preparation methods are provided in the SOP. The LC/MS/MS analytical conditions are provided as an example because each laboratory employed different instrumentation and LC columns. Each laboratory optimized the LC/MS/MS operating conditions and confirmed the achievement of adequate separation for each PFAA analyte peak prior to analyzing the recovery test samples. The participants in the validation study conducted recovery tests according to the SOP described in the following sections.

SOP for Inter-laboratory Tests

Chemicals and reagents

(1) Sodium ascorbate.—Guaranteed reagent grade (FUJIFILM Wako Pure Chemicals, Tokyo, Japan).
(2) Purified water.—Purified water for PFOS/PFOA analysis (FUJIFILM Wako Pure Chemical).
(3) Methanol.—LC/MS grade (FUJIFILM Wako Pure Chemicals).
(4) 0.1% ammonia in methanol solution.—A mixture of 25% ammonia water (0.4 mL) and methanol (99.6 mL).
(5) 10 mM ammonium acetate solution.—Ammonium acetate (7.7 g) dissolved in purified water (100 mL), and diluted (100x) using purified water.
(6) Acetonitrile.—LC/MS grade (FUJIFILM Wako Pure Chemicals).
(7) Native PFAA standard stock solution mixture (2 mg/L each).—The stock solution contained 2 mg/L each of

the linear isomer of the following 21 PFAAs: perfluorobutanesulfonic acid (PFBS), perfluoropentanesulfonic acid (PFPeS), perfluoroheptanesulfonic acid (PFHpS), PFHxS, PFOS, perfluorononanesulfonic acid (PFNS), perfluorodecanesulfonic acid (PFDS), perfluorododecanesulfonic acid (PFDoDS), perfluorobutanesulfonic acid (PFBA), perfluoropentanoic acid (PFPeA), perfluorohexanoic acid (PFHxA), perfluoroheptanoic acid (PFHpA), PFOA, perfluorononanoic acid (PFNA), perfluorodecanoic acid (PFDA), perfluoroundecanoic acid (PFUnDA), perfluorododecanoic acid (PFDoDA), perfluorotridecanoic acid (PFTrDA), perfluorotetradecanoic acid (PFTeDA), perfluorohexadecanoic acid (PFHxDA), and perfluorooctadecanoic acid (PFOcDA). For example, PFAC-MXC stock solution (Wellington Laboratories Inc., Ontario, Canada).

(8) ^{13}C-labeled PFAA extraction standard stock solution mixture (^{13}C-PFAA-ES, 2 mg/L each).—The stock solution contained 2 mg/L of the linear isomer of the following 13 ^{13}C-labeled PFAAs: $^{13}C_3$-PFBS, $^{13}C_3$-PFHxS, $^{13}C_8$-PFOS, $^{13}C_4$-PFBA, $^{13}C_5$-PFPeA, $^{13}C_5$-PFHxA, $^{13}C_4$-PFHpA, $^{13}C_8$-PFOA, $^{13}C_9$-PFNA, $^{13}C_6$-PFDA, $^{13}C_7$-PFUnDA, $^{13}C_2$-PFDoDA, and $^{13}C_2$-PFTeDA. For example, MPFAC-C-ES stock solution (Wellington Laboratories).

(9) ^{13}C-labeled PFAA injection standard stock solution mixture (^{13}C-PFAA-IS, 2 mg/L each).—The stock solution contained 2 mg/L each of the linear isomer of the following 4 ^{13}C-labeled PFAAs: $^{13}C_4$-PFOS, $^{13}C_3$-PFBA, $^{13}C_2$-PFOA, and $^{13}C_2$-PFDA. For example, MPFAC-C-IS stock solution (Wellington Laboratories).

(10) Native PFAA standard solution mixture A (0.1 mg/L each).—Dilute 500 µL of the native PFAA standard stock solution mixture (2 mg/L each) to 10 mL in a glass volumetric flask with methanol. The solution was distributed by the NIHS.

(11) Native PFAA standard solution mixture B (0.01 mg/L each).—Add 100 µL of the native PFAA standard solution mixture A (0.1 mg/L each) to a glass vial, and then add 900 µL of methanol.

(12) ^{13}C-PFAA-ES solution mixture (0.1 mg/L each).—Dilute 500 µL of the ^{13}C-PFAA-ES stock solution mixture (2 mg/L each) to 10 mL in a glass volumetric flask with methanol. The solution was distributed by the NIHS.

(13) ^{13}C-PFAA-IS solution mixture (0.1 mg/L each).—Dilute 500 µL of the ^{13}C-PFAA-IS stock solution mixture (2 mg/L each) to 10 mL in a glass volumetric flask with methanol. The solution was distributed by the NIHS.

Apparatus and instruments

(1) Sample collection bottles.—Glass bottles with a volume of 1,000–2,000 mL.—Wash with acetone and dry before use. Do not use that with fluorocarbon resin, fluoroplastic, or fluorocarbon polymer-coated bottle caps. Polypropylene bottles with a volume of 1,000–2,000 mL.—Wash with methanol and dry before use.

(2) Volumetric flasks.—Glass flasks. Wash with acetone and dry before use.

(3) Pipettes.—Pipettes with attachable polypropylene tips and whose volume are adjustable in the range of 20–250 µL.

(4) Test tubes.—Glass test tubes with a volume of 10 mL. Wash with acetone and dry before use.

(5) SPE columns.—SPE columns with a weak anion-exchange polymer packing content of ≥ 225 mg. For example, Oasis WAX Plus Short Cartridge (225 mg, Waters), InertSep mini MA-2 (280 mg, GL Sciences Inc.), or other SPE columns with equivalent or superior performance.

(6) SPE apparatus.—Vacuum manifold system with attachable SPE columns and adjustable pressure, or an automatic SPE device with attachable SPE columns and a constant-flow configuration for sample loading into the SPE column.

(7) Liquid chromatograph-tandem mass spectrometer (LC-MS/MS)

a. LC column.—A stainless steel tube with a 2.1–4.6 mm I.D. and a 7.5–25 cm length packed with silica gel (≤ 3 µm pore size) or another LC column with equivalent or superior performance.

b. Mobile phase.—Optimized mobile phase. For example, use 10 mM aqueous ammonium acetate solution as phase A and acetonitrile as phase B.

c. Mobile phase gradient.—Optimize the mobile phase conditions on the basis of peak separation between each PFAA analyte.

d. Ionization method.—Electrospray ionization (ESI) method in the negative-ion mode.

e. Detector conditions.—Selected ion monitoring (SIM), selected reaction monitoring (SRM) or other methods with an equivalent or superior performance.

Preparation of PFAA recovery test samples

(1) Collect tap water in a sampling bottle.

(2) Add approximately 10–20 mg of sodium ascorbate to 100 mL of tap water, then stir the mixture to facilitate dechlorination.

(3) Prepare PFAA-fortified tap water samples with each PFAA concentration of 1 and 10 ng/L as well as blank (unfortified) tap water and operation blank samples according to the procedures described below. Prepare 5 replicates of each PFAA-fortified tap water sample and 3–5 blank tap water and operation blank samples.

- PFAA-fortified tap water sample at 10 ng/L each PFAA.— Dispense 500 mL of dechlorinated tap water into a glass bottle using a graduated cylinder. Add 50 µL of native PFAA standard solution mixture A to the tap water, and then stir the solution.
- PFAA-fortified tap water sample at 1 ng/L each PFAA.— Dispense 500 mL of dechlorinated tap water into a glass bottle using a graduated cylinder. Add 50 µL of native PFAA standard solution mixture B to the tap water, and then stir the solution.
- Blank (unfortified) tap water samples.—Prepare 500 mL of dechlorinated tap water.
- Operation blank samples.—Prepare 500 mL of purified water.

Do not use containers comprising or apparatus containing fluorocarbon resin in any of the procedures.

Sample pretreatment

(1) Add 50 µL of ^{13}C-PFAA-ES standard solution mixture to each of 1 ng/L PFAA-fortified tap water, 10 ng/L PFAA-fortified tap water, blank tap water, and operation blank samples, and then stir the prepared samples.

(2) To condition the SPE column, add 4 mL of the 0.1% ammonia water/methanol solution, 4 mL of methanol, and 4 mL of purified water to the column.

(3) Load water sample onto the SPE column at a flow rate of 5 mL/min.

(4) Wash the SPE column with 5 mL of purified water, and then dispose the used water.

(5) Dry the SPE column by blowing N_2 gas or air through the column.

(6) Flush the column with 5 mL of the 0.1% ammonia water/methanol solution from the side opposite the loading section, and collect the effluent in a test tube.

(7) Concentrate the effluent to a volume of less than 0.45 mL by blowing N_2 gas or air above the solution, and then add 50 µL of the ^{13}C-PFAA-IS standard solution mixture to this.

(8) Adjust the total volume of the sample to 0.5 mL with methanol, and subject the sample to LC/MS/MS analysis.

Do not use containers comprising or apparatus containing fluorocarbon resin in any of the procedures.

Preparation of standard samples for calibration curves of each PFAA

Prepare the standard samples for the construction of calibration curves for each PFAA (STD0–STD6) by adding the specified amounts of native PFAA, ^{13}C-PFAA-ES, and ^{13}C-PFAA-IS standard mixtures, and methanol to vials for LC/MS/MS analysis. (**Table 2**).

Do not use containers comprising or apparatus containing fluorocarbon resin in any of the procedures.

LC/MS/MS analysis

(1) Inject a specific amount of the sample solutions and calibration curve standard samples prepared above to the LC/MS/MS instrument.

Repeat each analysis 3–5 times for the calibration curve standard samples, and then evaluate the repeatability and linearity of the calibration curves for each PFAA.

(2) Assign the PFAA peaks by comparing their retention times to those of the corresponding standard samples, and then measure the area of each PFAA peak.

Measure only the peak area of the linear isomer of each PFAA for the method validation study.

(3) Prepare the calibration curves at each PFAA concentration using the ratio of the peak area of each PFAA analyte to that of the corresponding ^{13}C-PFAA-ES or ^{13}C-PFAA-IS standard (**Table 3**).

(4) Calculate the PFAA analyte concentrations of the PFAA-fortified tap water samples in the eluent using the ratio of the peak area of each PFAA analyte to that of the corresponding ^{13}C-PFAA-ES or ^{13}C-PFAA-IS standard. If necessary, subtract the peak area of each PFAA in the blank tap water sample from that in the PFAA-fortified tap water sample prior to calculation.

(5) Divide each calculated PFAA analyte concentration of the PFAA-fortified samples in the eluent by the concentration ratio (1,000x) used in the sample pretreatment step to calculate the PFAA concentration of the water samples.

RESULTS AND DISCUSSION

Development of LC/MS/MS analytical conditions

The retention times and monitored ions of each native PFAAs and ^{13}C-PFAA examined in this study are listed in **Table 3**. The hydrogen-detached $[M\text{-}H]^-$ ion species, which were detected in the negative-ion mode, were selected as the precursor ions for all PFAA analytes. Several patterns were

Table 2 Preparation method of each standard solution used for the construction of calibration curves STD0–STD6.

	PFAA conc. (μg/L)	Added volume (μL)					Total volume (μL)
		STD mix.A (100 μg/L)	STD mix.B (10 μg/L)	ES STD mix. (100 μg/L)	IS STD mix. (100 μg/L)	Methanol	
STD0	0	-	-	50	50	400	500
STD1	1	-	50	50	50	350	500
STD2	2	-	100	50	50	300	500
STD3	5	-	250	50	50	150	500
STD4	10	50	-	50	50	350	500
STD5	20	100	-	50	50	300	500
STD6	50	250	-	50	50	150	500

observed for the product ions by LC/MS/MS analysis. For the PFSA analytes, the ions at m/z = 80 (SO_3^-) and 99 ($F+SO_3^-$) exhibit the highest intensity in all case. However, for the PFCA species, the product ions with the highest intensity varied among the analytes. Ions with m/z = 169 (C_3F_7), 219 (C_4F_9), and 269 (C_5F_{11}), which are fragmentation ions of the carbon chain, exhibited the highest intensities.

LC/MS/MS chromatograms of all PFAA analytes obtained under mobile-phase gradient conditions are shown in **Fig. 2**. Peaks assigned to the PFSA analytes were observed from 10.07 min (PFBS) to 21.26 min (PFDoDS), whereas those corresponding to the PFCA analytes were observed from 4.55 min (PFBA) to 24.39 min (PFOcDA) and spaced at almost equal intervals.

Sample pretreatment and LC/MS/MS analytical conditions of each laboratory

The sample pretreatment and LC/MS/MS analytical conditions developed by the NIHS (Organization A) and the 15 other laboratories (B–P, in random order) participating in the inter-laboratory test are shown in **Table 4**. Fourteen laboratories used the Oasis WAX Plus Short Cartridge (225 mg, Waters) for the SPE column, whereas two laboratories used the InertSep mini MA-2 column (280 mg, GL Sciences). In the SPE procedure, 12 laboratories used an automatic SPE device, such as a Sep-Pak concentrator (Waters, MA, USA), AQUALoader (GL Sciences), or an AquaTrace apparatus (GL Sciences), whereas 4 laboratories conducted SPE manually using a vacuum manifold. Four laboratories (G, K, M, and P) washed surface of the container used for the water samples with methanol and purified water after SPE column loading, and then loaded the washing liquid to SPE column to collect any PFAA analytes absorbed on the surface of the containers. Laboratories used LC columns from a variety of manufacturers; however, all 16 laboratories used ODS (C18) columns. Fifteen laboratories used a mobile-phase gradient of 10 mM aqueous ammonium acetate and acetonitrile, whereas one laboratory (P) used a gradient comprising 0.1% aqueous formic acid and 0.1% aqueous formic acid-methanol. All laboratories confirmed that adequate peak separation was achieved for each PFAA analyte. The laboratories used an LC injection volume of 1–5 μL depending on the capability of the instrument at hand. A retention gap column was used by 7 laboratories to reduce the blank level of PFAAs originating from the LC instrument. For mass spectrometer, seven laboratories used a SCIEX instrument, four each used Shimadzu Corporation and Waters instrument, and one laboratory used an Agilent instrument.

Validity assessment of calibration curves of each laboratory

The validity of the calibration curves must be ensured to accurately determine the results of the recovery test samples. Therefore, the validity of the calibration curve of each PFAA analyte was evaluated in the inter-laboratory test. The Japanese guideline for method validation of drinking water analyses [20] provides target values for trueness and repeatability at each calibration point. The target value for trueness is 80–120%, while for repeatability (RSD) is ≤ 20%. The validity of the calibration curve was confirmed in the 1–50 μg/L concentration range (STD1–STD6) provided by the SOP. The upper limit of the calibration curve was reduced to establish a narrower concentration range (e.g., 1–20 μg/L for STD1–STD5, or 1–10 μg/L for STD1–STD4) if the validity was not confirmed. Consequently, calibration curves were established in the concentration ranges of 1–50 μg/L (STD1–STD6) or 1–20 μg/L (STD1–STD5) in nearly all the laboratories. Moreover, all analytes were detectable at a level of 1 μg/L by all laboratories. It is imperative to confirm calibration curve linearity and to optimize LC/MS/

Table 3 Retention times and monitored ions of each PFAA analyte and ^{13}C-PFAA.

#	Analyte	Class	ES#	IS#	Monoisotopic mass	Retention time (min)	Precursor ion (*m/z*)	Ion species	Quantifier ion (*m/z*)	Qualifier ion1 (*m/z*)	Qualifier ion2 (*m/z*)
1	PFBS	Target	22	35	299.95	10.07	299	$[M-H]^-$	80	99	-
2	PFPeS	Target	22	35	349.95	12.00	349	$[M-H]^-$	80	99	-
3	PFHxS	Target	23	35	399.94	13.60	399	$[M-H]^-$	80	99	-
4	PFHpS	Target	23	35	449.94	15.10	449	$[M-H]^-$	80	99	-
5	PFOS	Target	24	35	499.94	16.49	499	$[M-H]^-$	80	99	-
6	PFNS	Target	24	35	549.93	17.78	549	$[M-H]^-$	80	99	-
7	PFDS	Target	24	35	599.93	19.04	599	$[M-H]^-$	80	99	-
8	PFDoDS	Target	24	35	699.92	21.26	699	$[M-H]^-$	80	99	-
9	PFBA	Target	25	36	213.99	4.55	213	$[M-H]^-$	169	-	-
10	PFPeA	Target	26	36	263.98	7.42	263	$[M-H]^-$	219	-	-
11	PFHxA	Target	27	36	313.98	9.46	313	$[M-H]^-$	269	119	-
12	PFHpA	Target	28	37	363.98	11.22	363	$[M-H]^-$	319	169	-
13	PFOA	Target	29	37	413.97	12.67	413	$[M-H]^-$	369	169	-
14	PFNA	Target	30	37	463.97	14.12	463	$[M-H]^-$	419	219	169
15	PFDA	Target	31	38	513.97	15.51	513	$[M-H]^-$	469	269	219
16	PFUnDA	Target	32	38	563.96	16.82	563	$[M-H]^-$	519	269	219
17	PFDoDA	Target	33	38	613.96	18.05	613	$[M-H]^-$	569	269	319
18	PFTrDA	Target	33	38	663.96	19.27	663	$[M-H]^-$	619	169	369
19	PFTeDA	Target	34	38	713.95	20.46	713	$[M-H]^-$	669	169	319
20	PFHxDA	Target	34	38	813.95	22.61	813	$[M-H]^-$	769	219	319
21	PFOcDA	Target	34	38	913.94	24.39	913	$[M-H]^-$	869	219	319
22	$^{13}C_3$-PFBS	ES	-	-	302.96	10.10	302	$[M-H]^-$	80	99	-
23	$^{13}C_3$-PFHxS	ES	-	-	402.95	13.60	402	$[M-H]^-$	80	99	-
24	$^{13}C_8$-PFOS	ES	-	-	507.96	16.52	507	$[M-H]^-$	80	99	-
25	$^{13}C_4$-PFBA	ES	-	-	218.00	4.56	217	$[M-H]^-$	172	-	-
26	$^{13}C_5$-PFPeA	ES	-	-	269.00	7.40	268	$[M-H]^-$	223	-	-
27	$^{13}C_5$-PFHxA	ES	-	-	319.00	9.51	318	$[M-H]^-$	273	-	-
28	$^{13}C_4$-PFHpA	ES	-	-	367.99	11.22	367	$[M-H]^-$	322	169	172
29	$^{13}C_8$-PFOA	ES	-	-	422.00	12.72	421	$[M-H]^-$	376	172	223
30	$^{13}C_9$-PFNA	ES	-	-	473.00	14.11	472	$[M-H]^-$	427	223	172
31	$^{13}C_6$-PFDA	ES	-	-	519.99	15.51	519	$[M-H]^-$	474	223	-
32	$^{13}C_7$-PFUnDA	ES	-	-	570.99	16.79	570	$[M-H]^-$	525	270	169
33	$^{13}C_2$-PFDoDA	ES	-	-	615.97	18.08	615	$[M-H]^-$	570	169	269
34	$^{13}C_2$-PFTeDA	ES	-	-	715.96	20.46	715	$[M-H]^-$	670	169	219
35	$^{13}C_4$-PFOS	IS	-	-	503.95	16.47	503	$[M-H]^-$	80	99	-
36	$^{13}C_3$-PFBA	IS	-	-	217.00	4.56	216	$[M-H]^-$	172	-	-
37	$^{13}C_2$-PFOA	IS	-	-	415.98	12.75	415	$[M-H]^-$	370	169	170
38	$^{13}C_2$-PFDA	IS	-	-	515.97	15.51	515	$[M-H]^-$	470	269	220

MS conditions including injection volume before undertaking PFAA quantification, because the linear range of the calibration curve depends on the performance and condition of the instrument used for the analysis.

According to the Japanese guidelines [22], carryover (*i.e.*, the residual analyte remaining in the instrument after a previous analysis) should also be evaluated. Some PFAA analytes were detected in the calibration curve blank samples by some laboratories. However, the peak areas of the blank samples were less than those of the STD1 samples. There-

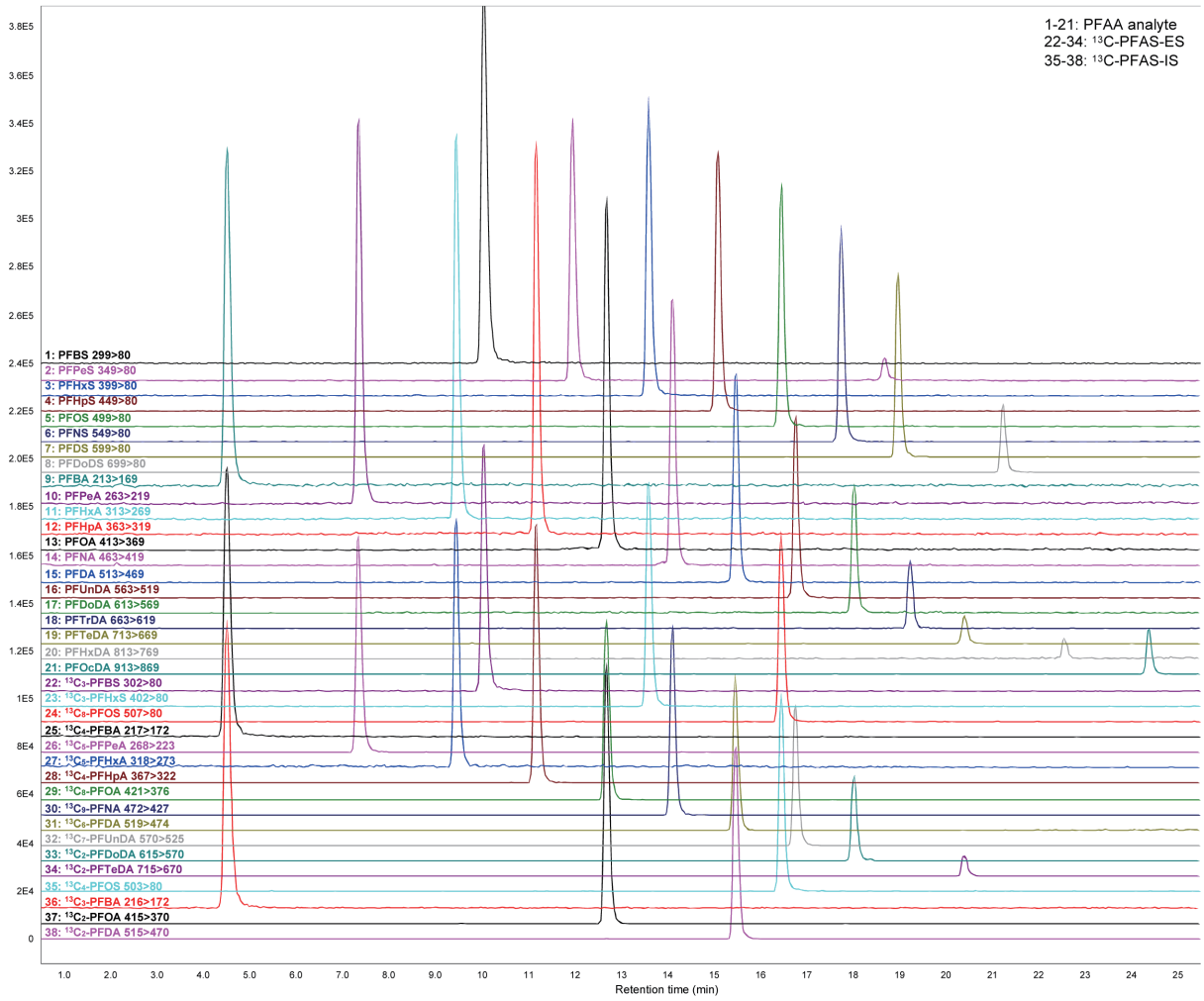


Fig. 2 LC/MS/MS chromatogram of each PFAA analytes, ^{13}C-PFAA-ES, and ^{13}C-PFAA-IS.

fore, the calibration curves and analytical results were not affected by the carryover of the PFAAs.

Validity assessment of recovery test water samples

The PFAA concentrations of the PFAA-fortified tap water and blank samples were qualified and quantified using the calibration curves established as described above. The PFAA concentrations in the operation blank samples were generally below the 1 ng/L LOQ. In instances where a PFAA analyte was detected in the blank sample, the concentration was below 2 ng/L. Therefore, each laboratory participating in the inter-laboratory test established method validation by controlling PFAA contamination in the laboratory environment.

PFOS, PFOA, and PFHxS were detected in the blank tap water samples by several laboratories at concentrations of up to 2.4, 8.1, and 1.7 ng/L, respectively. The presence of PFBA, PFPeA, PFHxA, PFHpA, and PFNA was also detected by several laboratories. The concentration of PFBA in the blank tap water samples, up to 13 ng/L, were higher than those of other PFAAs. Other PFAAs were not detected or were below the LOQ in most participating laboratories. If a PFAA analyte was detected in the blank tap water sample, the PFAA concentration of the blank sample was subtracted from that of the PFAA-fortified tap water sample prior to the calculation of recovery.

The recovery, repeatability (RSD_r), and reproducibility (RSD_R) of the PFAA analytes and ^{13}C-PFAA-ES are listed in **Tables 5 and 6**, respectively. All laboratories easily quantified the target PFAA at 1 ng/L in the PFAA-fortified tap water samples. The average recoveries of each PFAA analyte in all laboratories are shown in **Fig. 3**. According to Method 533 of the U.S. EPA, the average recovery of each analyte fortified at concentrations near or at the minimum reporting

Table 4a LC/MS/MS analytical conditions employed by the laboratories participating in the method validation test (1/2).

		Laboratory A	Laboratory B	Laboratory C	Laboratory D	Laboratory E	Laboratory F	Laboratory G	Laboratory H
Sampling	Location	Kawasaki city, Kanagawa prefecture	Hiroshima city, Hiroshima prefecture	Tsu city, Mie prefecture	Shinjuku city, Tokyo Metropolitan	Kawasaki city, Kanagawa prefecture	Osaka city, Osaka prefecture	Osaka city, Osaka prefecture	Hikigun, Saitama prefecture
SPE	Manufacturer	Waters	Waters	Waters	Waters	Waters	Waters	Waters	Waters
	Model	Oasis WAX Plus	Oasis WAX Plus	Oasis WAX Plus	Oasis WAX Plus	Oasis WAX Plus	Oasis WAX Plus	Oasis WAX Plus	Oasis WAX Plus
	Specification	225 mg	225 mg	225 mg	225 mg	225 mg	225 mg	225 mg	225 mg
SPE conditioning	Solution 1	0.1%Ammonia water methanol 4 mL	0.1%Ammonia water methanol 4 mL	0.1%Ammonia water methanol 4 mL	0.1%Ammonia water methanol 4 mL	0.1%Ammonia water methanol 4 mL	0.1%Ammonia water methanol 4 mL	0.1%Ammonia water methanol 4 mL	0.1%Ammonia water methanol 4 mL
	Solution 2	Methanol 4 mL	Methanol 4 mL	Methanol 4 mL	Methanol 4 mL	Methanol 4 mL	Methanol 4 mL	Methanol 4 mL	Methanol 4 mL
	Solution 3	Purified water 4 mL	Purified water 4 mL	Purified water 4 mL	Purified water 4 mL	Purified water 4 mL	Purified water 4 mL	Purified water 4 mL	Purified water 4 mL
Loading	Manufacturer	GL Sciences	Waters	-	Waters	Waters	Waters	Waters, GL Sciences	Waters
	Model	AquaTrace ASPE899	Sep-Pak concentrator SPC10-P	-	Sep-Pak concentrator Plus	Sep-Pak concentrator Plus	Sep-Pak concentrator Plus	Sep-Pak concentrator Plus, AQUALoaderIII	Sep-Pak concentrator Plus
	Flow rate	5 mL/min	5 mL/min	5 mL/min	5 mL/min	5 mL/min	5 mL/min	5 mL/min	5 mL/min
Wash	Solution	Purified water 5 mL	Purified water 5 mL	Purified water 5 mL	Purified water 5 mL	Purified water 5 mL	Purified water 5 mL	Purified water 5 mL	Purified water 5 mL
Elute	Solution	0.1%Ammonia in methanol 5 mL	0.1%Ammonia in methanol 5 mL	0.1%Ammonia in methanol 5 mL	0.1%Ammonia in methanol 5 mL	0.1%Ammonia in methanol 5 mL	0.1%Ammonia in methanol 5 mL	0.1%Ammonia in methanol 5 mL	0.1%Ammonia in methanol 5 mL
	Direction	Back flush	Back flush	Back flush	Back flush	Back flush	Back flush	Back flush	Back flush
LC	Manufacturer	Shimadzu	Shimadzu	Agilent	Waters	Shimadzu	Waters	Waters	Agilent
	Model	Prominence UFLC	Nexera X2	1260 InfintyII	UPLC I-Class	Nexera XR	UPLC H-Class	UPLC H-Class	1290 Infinity
LC column	Manufacturer	GL Sciences	Osaka Soda	GL Sciences	Waters	Waters	Waters	Merck	Merck
	Model	InertSustain AQ-C18	CAPCELL PAK C18	InertSustain C18	ACQUITY UPLC BEH C18	ACQUITY UPLC BEH C18	ACQUITY UPLC BEH C18	Ascentis C18	Ascentis C18
	Length	150 mm	150 mm	150 mm	100 mm	100 mm	100 mm	150 mm	150 mm
	I.D.	2.1 mm	2.0 mm	2.1 mm	2.1 mm	2.1 mm	2.1 mm	2.1 mm	2.1 mm
	Pore size	3 µm	3 µm	3 µm	1.7 µm	1.7 µm	1.7 µm	5 µm	3 µm
Retention gap column	Manufacturer	-	-	-	-	-	Waters	Merck	-
	Model	-	-	-	-	-	ACQUITY UPLC BEH C18	Ascentis C18	-
	Length	-	-	-	-	-	50 mm	50 mm	-
	I.D.	-	-	-	-	-	3.0 mm	2.1 mm	-
	Pore size	-	-	-	-	-	1.7 µm	5 µm	-

Table 4a *Continued.*

		Laboratory A	Laboratory B	Laboratory C	Laboratory D	Laboratory E	Laboratory F	Laboratory G	Laboratory H
LC condition	Mobile phase A	10 mM ammonium acetate solution	10 mM ammonium acetate solution	0.1% formic acid and 0.5 mM ammonium acetate solution	10 mM ammonium acetate solution	10 mM ammonium acetate solution	10 mM ammonium acetate solution	10 mM ammonium acetate solution	10 mM ammonium acetate solution
	Mobile phase B	Acetonitrile	Acetonitrile	Acetonitrile	90%Acetonitrile and 10%10 mM ammonium acetate solution	Acetonitrile	Acetonitrile	Acetonitrile	Acetonitrile
	Mobile phase condition (B%)	25% (0–1 min)–100% (26–30 min)	20% (0–2 min)–95% (22–28 min)	20% (0–2 min)–95% (22–28 min)	40% (0–1 min)–100% (10–15 min)	20% (0–2 min)–95% (22–28 min)	5% (0 min)–95% (10–15 min)	20% (0–2 min)–95% (22–28 min)	20% (0–2 min)–95% (18–23 min)
	Flow rate	0.2 mL/min	0.2 mL/min	0.2 mL/min	0.2 mL/min	0.2 mL/min	0.3 mL/min	0.2 mL/min	0.3 mL/min
	Column temp.	40°C	40°C	40°C	40°C	40°C	40°C	40°C	40°C
	Sample temp.	5°C	15°C	10°C	10°C	5°C	9°C	10°C	4°C
	Injection vol.	5 µL	5 µL	2 µL	2 µL	5 µL	2 µL	5 µL	5 µL
MS	Manufacturer	Shimadzu	Shimadzu	Sciex	Waters	Sciex	Waters	Waters	Sciex
	Model	LCMS-8050	LCMS-8040	QTRAP5500	Xevo TQ	QTRAP4500	Xevo TQ-XS	Xevo TQS-micro	QTRAP 4500
MS condition	Ionization	ESI Negative	ESI Negative	ESI Negative	ESI Negative	ESI Negative	ESI Negative	ESI Negative	ESI Negative
	Ionization voltage	-3.5 kV	-3.5 kV	−4.5 kV	-1.0 kV	-4.5 kV	-0.5kV	-1.0 kV	-4.5 kV
	Other MS parameters	Nebulizer gas flow 3 L/min Drying gas flow 10 L/min Heating gas flow 10 L/min Interface temp. 300°C Desolvent temp. 500°C DL temperature 250°C Heat block temp. 400°C Conversion dynode voltage 10 kV CID gas pressure 270 kPa	Nebulizer gas flow 1.5 L/min Drying gas flow 10 L/min Heat block temp. 400°C DL temperature: 250°C	Curtain gas 30 psi Collision gas 9 Ion source temp. 300°C Ion source gas 1 80 psi Ion source gas 2 80 psi	Desolvent gas flow 1100 L/h Corn gas flow 50 L/h Desolvent temp. 500°C Ion source temp. 150°C	Curtain gas 30 psi Collision gas 12 Ion source temp. 400°C Ion source gas 1 30 psi Ion source gas 2 30 psi	Desolvent gas flow 1200 L/h Corn gas flow 150 L/h Desolvent temp. 450°C Ion source temp. 120°C	Desolvent gas flow 1100 L/h Corn gas flow 50 L/h Desolvent temp. 500°C Ion source temp. 150°C	Curtain gas 20 psi Collision gas 10 Ion source temp. 300°C Ion source gas 1 70 psi Ion source gas 2 60 psi

Table 4b LC/MS/MS analytical conditions employed by the laboratories participating in the method validation test (2/2).

		Laboratory I	Laboratory J	Laboratory K	Laboratory L	Laboratory M	Laboratory N	Laboratory O	Laboratory P
Sampling	Location	Fukuyama city, Hiroshima prefecture	Chigasaki city, Kanagawa prefecture	Kouzagun, Kanagawa prefecture	Chiba city, Chiba prefecture	Gifu city, Gifu prefecture	Iruma city, Saitama prefecture	Hachioji city, Tokyo Metropolitan	Nagoya city, Aichi prefecture
SPE	Manufacturer	Waters	Waters	Waters	Waters	Waters	GL Sciences	GL Sciences	Waters
	Model	Oasis WAX Plus	Oasis WAX Plus	Oasis WAX Plus	Oasis WAX Plus	Oasis WAX Plus	InertSep mini MA-2	InertSep mini MA-2	Oasis WAX Plus
	Specification	225 mg	225 mg	225 mg	225 mg	225 mg	280 mg	280 mg	225 mg
SPE conditioning	Solution 1	0.1%Ammonia water methanol 4 mL	0.1%Ammonia water methanol 4 mL	0.1%Ammonia water methanol 4 mL	0.1%Ammonia water methanol 4 mL	0.1%Ammonia water methanol 4 mL	0.1%Ammonia water methanol 4 mL	0.1%Ammonia water methanol 4 mL	0.1%Ammonia water methanol 4 mL
	Solution 2	Methanol 4 mL	Methanol 4 mL	Methanol 4 mL	Methanol 4 mL	Methanol 4 mL	Methanol 4 mL	Methanol 4 mL	Methanol 4 mL
	Solution 3	Purified water 4 mL	Purified water 4 mL	Purified water 4 mL	Purified water 4 mL	Purified water 4 mL	Purified water 4 mL	Purified water 4 mL	Purified water 4 mL
Loading	Manufacturer	-	Waters	Waters	Waters	-	GL Sciences	GL Sciences	
	Model	-	Sep-Pak concentrator Plus	Sep-Pak concentrator Plus	Sep-Pak concentrator Plus	-	AquaTrace ASPE899	AQUALoader	
	Flow rate	5 mL/min	5 mL/min	5 mL/min	5 mL/min	5 mL/min	5 mL/min	5 mL/min	5 mL/min
Wash	Solution	Purified water 5 mL	Purified water 5 mL	Purified water 5 mL	Purified water 5 mL	Purified water 5 mL	Purified water 5 mL	Purified water 5 mL	Purified water 5 mL
Elute	Solution	0.1%Ammonia in methanol 5 mL	0.1%Ammonia in methanol 5 mL	0.1%Ammonia in methanol 5 mL	0.1%Ammonia in methanol 5 mL	0.1%Ammonia in methanol 5 mL	0.1%Ammonia in methanol 5 mL	0.1%Ammonia in methanol 5 mL	0.1%Ammonia in methanol 5 mL
	Direction	Back flush	Back flush	Back flush	Back flush	Back flush	Back flush	Back flush	Back flush
LC	Manufacturer	Sciex	Waters	Shimadzu	Shimadzu	Agilent Technologies	Shimadzu	Agilent	Shimadzu
	Model	ExionLC	UPLC H-Class	Nexera X2	Prominence UFLC	1260 Infinity II binary pump	Nexera UFLC	1260 Infinity II Prime	Prominence UFLC
LC column	Manufacturer	Waters	Waters	Waters	Osaka Soda	YMC	GL Sciences	Agilent	Waters
	Model	ACQUITY UPLC BEH C18	ACQUITY UPLC BEH C18	ACQUITY UPLC BEH C18	CAPCELL PAK C18 MGII	YMC-Triart C18	InertSustain C18 HP	Zorbax Eclipse plus C18 RRHD	X bridge C18
	Length	100 mm	100 mm	100 mm	150 mm	150 mm	150 mm	100 mm	50 mm
	I.D.	2.1 mm	2.1 mm	2.1 mm	2.0 mm	2.1 mm	2.1 mm	2.1 mm	1.0 mm
	Pore size	1.7 µm	1.7 µm	1.7 µm	3 µm	5 µm	3 µm	1.8 µm	3.5 µm
Retention gap column	Manufacturer	-	Waters	-	-	YMC	GL Sciences	Agilent	Osaka Soda
	Model	-	ACQUITY UPLC BEH C18	-	-	YMC-Triart C18	Delay column for PFAS	Zorbax Eclipse plus C18	CAPCELL PAC DD
	Length	-	50 mm	-	-	50 mm	30 mm	50 mm	35 mm
	I.D.	-	3.0 mm	-	-	2.1 mm	3.0 mm	4.6 mm	1.5 mm
	Pore size	-	1.7 µm	-	-	5 µm	-	3.5 µm	5 µm

Table 4b *Continued*.

		Laboratory I	Laboratory J	Laboratory K	Laboratory L	Laboratory M	Laboratory N	Laboratory O	Laboratory P
LC condition	Mobile phase A	10 mM ammonium acetate solution	10 mM ammonium acetate solution	10 mM ammonium acetate solution	10 mM ammonium acetate solution	10 mM ammonium acetate solution	10 mM ammonium acetate solution	2 mM ammonium acetate solution	0.1% formic acid
	Mobile phase B	Acetonitrile	Acetonitrile	Acetonitrile	Acetonitrile	Acetonitrile	Acetonitrile	Acetonitrile	0.1% formic acid in Methanol
	Mobile phase condition (B%)	5% (0 min)–98% (8.25–11 min)	1% (0–0.5 min)–30% (0.5–5 min)–65% (8–8.5 min)–95% (9.5–12 min)	20% (0–2 min)–95% (22–28 min)	25% (0–1 min)–95% (26–30 min)	25% (0 min)–40% (1 min) − 95% (10–14 min)	20% (0-2 min)–100%(15 min)	5% (0 –0.5 min)–30% (2 min)–100% (19–21 min)	1% (0–3 min)–95% (25–28 min)
	Flow rate	0.6 mL/min	0.35 mL/min	0.2 mL/min	0.2 mL/min	0.25 mL/min	0.3mL/min	0.3 mL/min	0.2 mL/min
	Column temp.	40°C	40°C	40°C	40°C	40°C	40°C	40°C	40°C
	Sample temp.	10°C	5°C	10°C	10°C	5°C	5°C	4°C	4°C
	Injection vol.	5 µL	2 µL	5 µL	5 µL	1 µL	1 µL	2 µL	5 µL
MS	Manufacturer	Sciex	Waters	Shimadzu	Sciex	Sciex	Sciex	Agilent	Shimadzu
	Model	QTrap4500	Xevo TQS-micro	LCMS8050	TQ4500	QTRAP5500	4000 QTRAP	Ultivo	LCMS-8050
MS condition	Ionization	ESI Negative	ESI Negative	ESI Negative	ESI Negative	ESI Negative	ESI Negative	ESI Negative	ESI Negative
	Ionization voltage	-4.5kV	-0.5 kV	-3.0 kV	-4.4 kV	-3.5 kV	-4500V	-2.5 kV	-3.5 kV
	Other MS parameters	Curtain gas 25 psi Collision gas 10 Ion source temp.400°C Ion source gas 1 50 psi Ion source gas 2 80 psi	Desolvent gas flow 700 L/h Corn gas flow 70 L/h Desolvent temp. 550°C Ion source temp. 120°C	Nebulizer gas flow 2 L/min Drying gas flow 10 L/mi Heating gas flow 10 L/min Interface temp. 200°C DL temperature 200°C Heat block temp. 400°C	Curtain gas 10 psi Collision gas 8 Ion source temp. 450°C Ion source gas 1 70 psi Ion source gas 2 40 psi	Curtain gas 30 psi Collision gas 7 Ion source temp. 400°C Ion source gas 1 60 psi Ion source gas 2 50 psi	Curtain gas 20 psi Collision gas 12 Ion source temp. 400°C Ion source gas 1 50 psi Ion source gas 2 30 psi	Desolvent gas flow 10 L/min Sheath gas flow 11 L/min Desolvent temp. 250°C Sheath gas temp. 300°C	Nebulizer gas flow 3 L/min Drying gas flow 10 L/min Heating gas flow 10 L/min Interface temp. 300°C DL temperature 250°C Heat block temp. 400°C Conversion dynode voltage 10 kV CID gas pressure 270 kPa

Table 5 Recovery, repeatability (RSD_r), and reproducibility (RSD_R) of each PFAA analyte.

#	Analyte	Conc. (ng/L)	Recovery (%)																RSD_r (%)														RSD_R (%)
			A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	
1	PFBS	1	102	75	76	63	83	75	114	104	59	84	87	95	100	104	82	98	22	24	10	64	12	24	4	12	16	2	9	9	5	10	9	2	18
		10	106	101	111	123	79	70	105	90	62	80	85	111	99	102	85	102	14	5	5	28	7	12	5	3	4	7	6	5	5	4	6	3	18
2	PFPeS	1	76	105	97	64	83	77	100	102	57	85	82	94	76	117	69	82	15	11	5	12	11	12	5	4	11	5	6	9	5	3	3	8	19
		10	82	100	124	72	79	66	100	97	74	80	80	107	95	109	82	91	14	12	4	18	7	10	7	3	2	6	4	10	6	2	2	4	17
3	PFHxS	1	80	96	85	47	92	75	106	103	76	95	97	106	90	101	77	91	16	13	9	26	12	15	5	6	6	4	13	10	7	3	6	5	17
		10	81	92	112	64	83	67	100	93	78	87	92	95	95	99	85	98	16	10	5	20	8	10	7	3	5	7	6	4	2	5	3	2	14
4	PFHpS	1	78	108	77	85	86	123	103	87	66	85	80	102	86	90	71	84	18	23	6	13	12	7	8	5	4	2	10	8	5	5	7	8	17
		10	80	116	99	59	84	71	99	90	85	84	78	93	96	99	85	108	17	7	7	20	6	9	7	5	6	6	8	5	3	5	3	9	16
5	PFOS	1	50	107	74	23	86	78	102	77	82	77	100	73	103	89	55	89	32	11	11	45	21	10	9	27	12	10	11	3	6	19	6	11	27
		10	61	111	98	49	75	65	100	74	92	74	87	81	99	88	77	117	17	5	4	16	8	12	8	17	5	6	6	3	4	6	3	5	22
6	PFNS	1	50	119	73	36	80	47	94	47	75	50	83	65	85	56	96	86	17	10	3	24	8	23	6	38	3	13	5	6	5	7	2	1	32
		10	41	96	102	48	55	46	93	53	82	63	88	76	94	52	73	98	15	6	6	27	9	16	4	22	2	6	8	4	4	5	4	11	29
7	PFDS	1	56	83	104	54	62	50	95	41	72	38	73	64	80	41	89	69	15	16	4	25	8	13	6	38	7	16	6	7	5	6	5	11	30
		10	33	85	91	49	42	36	90	42	69	59	80	59	96	46	64	77	16	12	7	12	9	18	6	24	7	7	8	5	2	8	10	8	33
8	PFDoDS	1	22	68	78	52	61	43	95	32	41	55	63	63	81	34	60	72	17	20	6	15	8	2	9	34	6	9	13	4	3	14	16	4	34
		10	24	70	61	42	36	31	86	34	44	56	68	44	88	39	36	99	19	11	7	22	9	2	7	20	6	7	10	4	4	6	32	12	42
9	PFBA	1	N.A.	96	90	73	90	93	68	108	102	69	91	67	99	56	70	101	N.A.	34	40	83	21	7	26	9	14	28	17	11	7	9	14	31	19
		10	60	153	84	85	96	102	92	93	77	80	98	100	101	55	94	110	35	7	7	43	2	1	9	2	7	7	5	11	1	2	5	2	24
10	PFPeA	1	63	92	81	30	101	63	61	104	100	118	83	99	99	91	99	116	25	5	20	84	9	8	20	4	11	4	11	6	4	10	5	13	26
		10	71	100	94	78	100	76	72	92	102	108	88	84	98	86	111	165	17	9	5	18	1	3	6	2	5	6	5	5	2	2	2	2	23
11	PFHxA	1	60	95	103	50	101	69	67	107	116	123	89	86	96	94	107	337	31	8	17	47	9	19	27	12	9	3	12	3	4	13	4	12	61
		10	68	96	116	59	102	91	70	92	104	111	100	87	97	84	114	325	20	9	5	13	3	0	7	3	6	6	5	6	2	6	3	11	56
12	PFHpA	1	57	101	98	44	77	61	76	104	70	95	98	82	106	96	74	95	32	4	15	36	28	48	11	9	15	6	5	12	4	10	6	14	22
		10	62	107	113	46	73	68	87	91	73	86	94	71	100	85	81	112	17	15	4	9	7	5	6	3	10	5	4	5	5	4	3	2	22
13	PFOA	1	51	101	95	50	49	34	74	106	93	90	98	104	103	103	73	113	44	10	15	26	44	53	41	5	13	4	15	15	6	8	6	18	30
		10	62	102	93	55	60	65	93	89	103	87	86	84	103	94	78	101	18	16	7	16	18	4	8	1	4	7	6	5	6	5	3	2	19
14	PFNA	1	55	91	68	162	88	87	92	99	166	82	94	84	102	97	71	105	20	23	9	11	24	17	16	13	8	11	10	11	6	11	7	21	30
		10	65	99	82	69	78	77	93	83	173	82	96	81	105	85	75	120	14	10	4	17	8	10	6	7	5	6	8	6	5	6	4	9	28
15	PFDA	1	37	71	78	51	72	64	97	69	67	65	89	84	103	64	65	156	28	2	4	12	16	5	8	25	11	8	5	4	8	7	8	33	35
		10	45	87	95	48	64	56	93	62	76	66	91	82	101	59	69	114	13	14	1	5	5	10	7	22	5	7	6	7	3	5	9	3	27
16	PFUnDA	1	27	64	66	45	81	44	109	48	67	58	85	74	93	47	75	96	18	12	7	21	8	14	18	30	3	13	6	8	9	10	6	16	33
		10	34	79	94	45	48	40	112	42	66	61	86	86	100	50	64	129	13	14	8	6	7	18	19	25	1	9	4	5	4	8	11	10	40
17	PFDoDA	1	25	66	66	39	45	56	98	45	67	61	82	56	90	36	68	157	27	8	5	8	17	8	10	31	3	9	6	4	6	7	7	28	47
		10	30	75	86	44	39	36	99	38	59	61	82	54	108	41	56	90	16	6	4	17	7	12	13	26	5	7	6	5	6	7	15	10	40
18	PFTrDA	1	34	77	47	39	40	55	111	38	67	43	91	56	104	31	59	79	15	10	14	14	16	4	9	34	6	10	6	4	5	8	13	11	42
		10	26	78	76	42	34	33	103	34	58	54	91	54	100	43	40	114	17	12	6	16	7	2	18	25	4	9	9	5	4	8	26	5	47
19	PFTeDA	1	14	72	43	32	44	29	112	30	52	50	70	17	99	34	46	130	16	7	8	12	17	4	12	34	9	12	7	28	4	5	30	27	62
		10	25	69	60	38	38	30	105	30	54	66	65	40	98	44	28	73	20	12	9	15	9	10	18	22	2	10	13	9	4	8	32	8	45
20	PFHxDA	1	58	74	12	29	44	27	112	20	70	47	38	7	98	54	31	82	13	11	14	23	21	2	11	22	4	9	26	28	5	6	68	6	61
		10	28	80	21	48	39	23	105	27	71	63	31	32	97	53	23	92	21	12	43	17	9	23	18	22	4	7	51	4	3	7	10	8	56
21	PFOcDA	1	18	78	14	30	52	20	114	14	74	50	12	8	104	35	37	89	48	5	3	25	19	8	12	31	6	7	116	29	8	10	41	12	74
		10	33	92	1	49	48	28	109	29	65	64	5	22	92	62	37	101	17	10	14	21	7	22	17	29	3	14	127	2	8	8	9	5	64

N.A.: Not available because of the negative value obtained by subtracting the concentration of the blank tap water sample from that of PFAA-added tap water sample.

Table 6 Recovery, repeatability (RSD_r), and reproducibility (RSD_R) of each ^{13}C-PFAA-ES.

#	^{13}C-PFAA-ES	Conc. (ng/L)	Recovery (%)																RSD_r (%)														RSD_R (%)
			A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	
1	$^{13}C_3$-PFBS	1	91	92	95	108	87	93	111	99	59	80	82	104	94	103	79	108	14	10	4	18	14	16	4	3	10	1	5	8	3	4	3	5	15
		10	93	97	106	137	85	80	114	93	62	80	83	105	93	103	79	100	14	4	8	26	9	8	4	2	6	6	5	5	2	6	3	4	18
2	$^{13}C_3$-PFHxS	1	69	78	90	63	92	91	102	101	79	83	87	104	94	103	79	106	15	11	5	22	17	15	2	2	8	1	6	11	2	6	3	7	14
		10	71	81	100	65	88	80	104	94	78	82	90	102	95	103	79	94	15	12	8	17	10	8	4	2	6	7	6	7	5	6	4	3	13
3	$^{13}C_8$-PFOS	1	53	89	82	54	82	83	99	85	95	63	90	80	95	75	81	122	16	7	4	18	16	12	3	14	5	5	4	6	2	5	3	15	20
		10	54	91	87	63	78	76	102	78	94	62	92	85	93	72	79	113	17	5	9	19	7	19	3	13	3	9	5	8	3	5	2	6	19
4	$^{13}C_4$-PFBA	1	60	88	80	98	99	99	80	96	81	86	86	93	96	60	85	107	12	8	4	14	3	3	4	2	5	2	6	3	2	2	3	10	15
		10	64	90	79	90	100	99	91	93	79	86	93	92	94	59	90	109	16	7	4	13	4	3	7	1	3	6	4	3	1	3	2	5	14
5	$^{13}C_5$-PFPeA	1	58	85	90	68	103	86	67	96	105	107	84	82	95	89	103	172	13	7	4	6	2	2	4	2	5	3	6	6	2	2	2	3	27
		10	61	86	89	72	106	85	74	93	104	107	93	86	94	87	107	162	16	8	3	8	3	3	8	1	4	7	5	4	3	3	2	4	24
6	$^{13}C_5$-PFHxA	1	55	91	114	58	107	91	63	98	131	105	91	85	96	94	55	312	14	7	5	9	4	3	4	2	5	3	5	4	3	3	14	40	58
		10	58	91	112	58	110	93	68	93	131	105	98	87	95	91	58	335	18	8	3	12	3	2	7	2	4	5	5	5	2	3	18	12	62
7	$^{13}C_4$-PFHpA	1	56	94	95	48	83	91	84	98	67	82	88	87	98	93	75	115	13	7	8	9	17	14	4	3	5	2	4	5	6	3	3	9	20
		10	54	92	102	47	78	80	91	92	68	81	92	82	95	89	79	114	16	8	8	10	9	3	7	2	6	7	4	7	5	5	3	2	20
8	$^{13}C_8$-PFOA	1	53	91	80	52	49	87	87	96	138	84	93	98	98	94	75	124	13	11	6	9	13	4	4	2	4	3	4	5	2	5	3	2	27
		10	55	89	84	53	50	80	95	90	144	82	96	93	95	93	76	113	14	11	5	15	14	2	6	2	2	8	3	5	3	3	2	2	27
9	$^{13}C_9$-PFNA	1	47	78	72	53	87	99	92	92	175	75	89	89	99	96	47	111	9	13	7	7	17	11	4	4	5	6	5	9	5	4	9	11	34
		10	54	78	76	55	80	87	98	84	180	73	90	89	95	88	54	99	14	10	8	15	10	11	6	5	3	9	5	4	4	3	14	10	34
10	$^{13}C_6$-PFDA	1	33	76	79	49	75	74	92	70	90	54	88	83	101	66	67	112	17	5	5	11	15	12	3	23	4	7	3	2	5	2	7	6	26
		10	40	75	87	44	71	73	97	62	86	55	90	87	96	62	68	99	13	10	5	9	8	20	5	20	4	9	6	4	4	2	9	2	25
11	$^{13}C_7$-PFUnDA	1	28	54	77	44	56	54	115	47	85	49	83	70	99	49	60	115	8	13	4	17	12	21	11	34	5	8	5	3	9	3	7	19	38
		10	31	55	85	42	51	54	117	41	77	49	88	77	97	50	63	112	13	10	6	5	9	22	18	23	5	12	5	6	4	6	10	14	38
12	$^{13}C_2$-PFDoDA	1	28	59	74	41	45	47	102	41	76	48	85	52	104	43	28	114	11	11	5	9	15	19	9	37	2	5	4	3	7	4	11	20	44
		10	26	63	81	42	41	46	104	36	63	47	87	63	99	44	26	113	16	4	6	21	10	18	10	22	6	9	6	3	5	7	16	17	46
13	$^{13}C_2$-PFTeDA	1	21	50	53	35	45	21	110	30	60	50	78	38	94	42	38	117	13	14	4	12	17	10	12	37	3	6	8	2	2	6	37	22	54
		10	22	54	58	36	40	34	112	27	44	47	69	54	96	44	28	108	17	10	6	17	10	6	18	22	6	8	14	3	4	7	33	16	51

level (within a factor of two times the minimum reporting level concentration) must be 50–150% of the true value, and the results for the analytes fortified at all other concentrations must be 70–130% of the true value [19]. The recovery gradually decreases for longer chain PFAAs. The average recoveries of PFDS, PFDoDS, PFUnDA, and other PFCAs with longer chains than PFUnDA were below 70%. The average recoveries of PFHxDA and PFOcDA were nearly 50%,

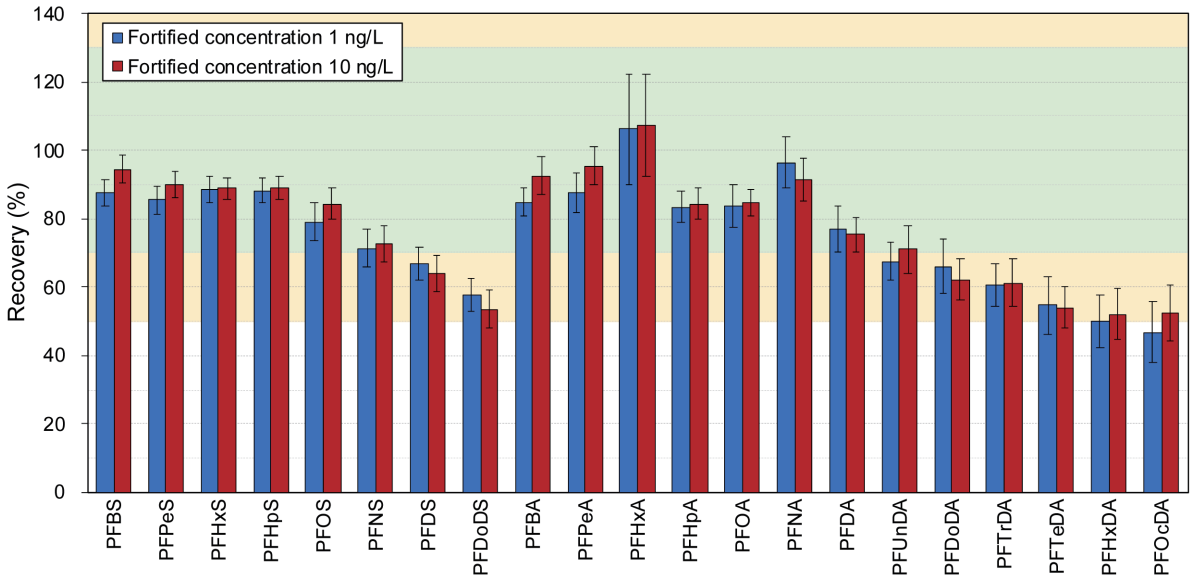


Fig. 3 Recoveries of each PFAA analyte in all laboratories (Average ± S.E.).

which does not satisfy the above-mentioned criteria.

The average recovery of each ^{13}C-PFAA-ES in all laboratories are shown in **Fig. 4**. According to Method 533 of the U.S. EPA, the percent recovery of each analogue must be within a range of 50–200% [19]. The recovery gradually decreases for longer chain PFAAs. However, the average recovery of all ^{13}C-PFAA-ES exceeded 50%, which satisfies the above-mentioned criteria.

The low recoveries of the long-chain PFAA analytes and ^{13}C-PFAA-ES are attributed to PFAA absorption on the surface of the containers during the SPE process [23,24]. If these losses are not corrected by the application of the recovery calculation using the ^{13}C-PFAA-ES, recovery is affected by the absorption. However, relatively good recovery was obtained by laboratories G, K, M, and P (**Tables 5 and 6**), which washed the container surfaces with methanol and purified water after SPE column loading. Therefore, it is advisable to wash container surface and then loaded the washing liquid to the SPE column to collect any PFAA analytes absorbed on the surface of the containers.

The recovery (trueness), RSD_r, and RSD_R of each PFAA analyte corrected by the ^{13}C-PFAA-ES are listed in **Table 7**. The average trueness for all PFAA analytes obtained in all laboratories are shown in **Fig. 5**. Good trueness is obtained when recovery is corrected by the ^{13}C-PFAA-ES. The average trueness for all PFAAs (except for PFDoDS) satisfied the 70–130% criterion of U.S.EPA Method 533 [19] and the Japanese guideline [22]. Although PFHxDA and PFOcDA satisfied the target value of trueness, wide variability is observed for these two PFCAs. The low trueness of PFDoDS is attributed to the correction of its recovery by $^{13}C_8$-PFOS (see **Table 3**). The recovery of $^{13}C_8$-PFOS (**Fig. 4**) differs from that of PFDoDS (**Fig. 3**), which causes the applied to PFDoDS to be inaccurate. The recoveries of PFHxDA and PFOcDA were corrected in an analogous manner by $^{13}C_2$-PFTeDA, which differs in its behavior from these two PFCAs. Further, the recoveries of PFHxDA and PFOcDA were nearly 50% or lower (**Fig. 3**). The low recoveries of these PFAAs may affect their wide variability.

The trueness of PFBA does not satisfy the target value in several laboratories, despite its short-chain length. This behavior is particularly evident at the lower concentrations (1 ng/L) in the PFAA-fortified tap water samples. Low trueness is observed here, because the PFBA concentrations of the blank tap water samples are high in several laboratories. Therefore, the subtraction of the blank PFBA concentration is not accurate. A negative value is even obtained in laboratory A for this reason.

The RSD_r of each PFAA corrected by ^{13}C-PFAA-ES satisfies the target value (≤ 20%) of the Japanese guideline [22] for almost all PFAAs (**Table 7**). On the other hand, the RSD_r of each PFAA without recovery correction by ^{13}C-PFAA-ES widely differ among the participating laboratories (**Table 5**).

The evaluation of RSD_R is essential in any inter-laboratory study. However, a target value of RSD_R is not provided in the Japanese guideline [22], although an intermediate precision value (≤ 25%) is stated. Generally, intermediate precision is less than RSD_R in physicochemical examination. Therefore,

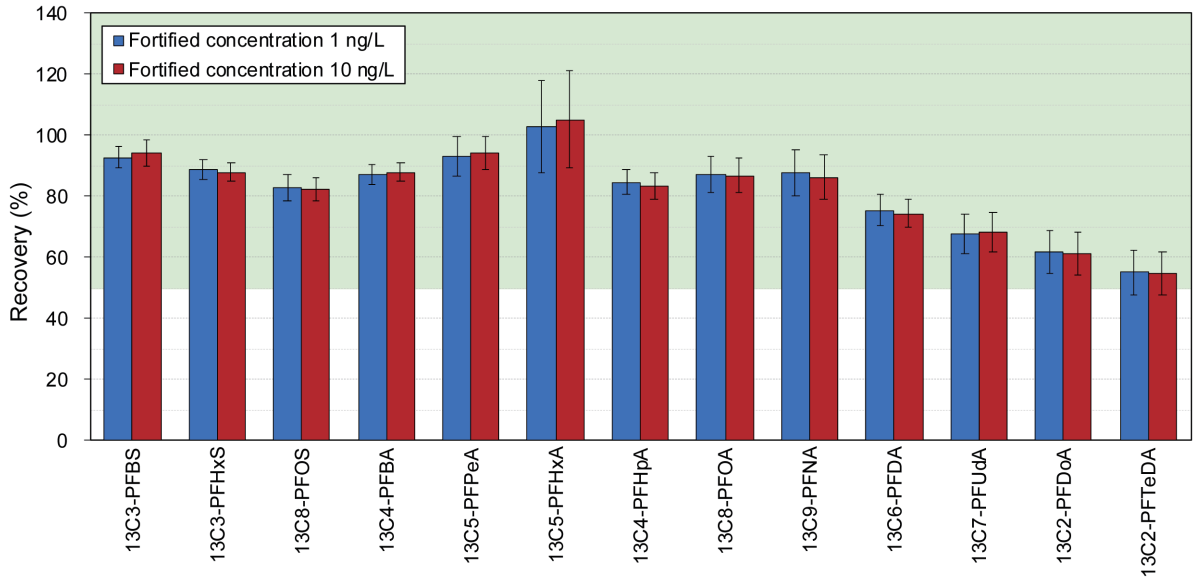


Fig. 4 Recoveries of each ^{13}C-PFAA-ES in all laboratories (Average ± S.E.).

Table 7 Trueness, repeatability (RSD_r), and reproducibility (RSD_R) of each PFAA analyte corrected by ^{13}C-PFAA-ES.

#	Analyte	Conc. (ng/L)	Trueness (%)																RSD_r (%)														RSD_R (%)
			A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	
1	PFBS	1	118	97	84	71	98	91	92	104	104	110	104	97	107	92	99	89	8	14	10	28	7	7	9	12	10	2	4	4	4	12	6	2	11
		10	118	107	109	99	95	96	89	99	100	103	104	109	105	100	104	104	2	4	4	6	4	4	2	2	2	9	2	5	4	4	4	2	7
2	PFPeS	1	111	107	107	61	98	86	88	98	126	109	101	86	90	95	95	86	6	18	3	9	4	6	2	2	3	5	2	8	7	6	2	4	15
		10	111	104	123	67	94	86	86	105	116	103	99	102	102	107	101	92	2	8	4	6	4	3	5	1	7	8	3	11	5	5	2	5	13
3	PFHxS	1	118	117	96	77	100	90	101	101	98	118	112	103	107	97	99	94	8	12	9	14	8	7	2	3	6	5	4	9	8	8	4	6	11
		10	114	115	113	102	95	91	94	99	102	105	103	96	100	97	106	106	1	8	3	5	5	2	4	1	3	9	2	6	4	4	1	3	7
4	PFHpS	1	112	137	84	111	93	123	113	84	108	114	94	87	105	93	120	95	3	20	3	21	6	4	6	4	9	2	6	13	5	9	4	3	15
		10	112	146	101	91	95	90	94	96	107	102	88	92	101	95	107	109	3	17	2	7	5	4	4	3	6	8	3	7	3	3	2	2	14
5	PFOS	1	106	105	93	52	94	82	99	95	89	176	101	95	105	102	83	98	6	11	9	26	17	18	9	15	5	9	7	3	6	16	7	8	25
		10	116	125	115	87	93	90	95	96	98	125	93	99	101	118	99	104	2	9	5	11	3	9	6	3	6	9	2	9	3	8	2	5	12
6	PFNS	1	84	120	112	74	98	57	106	67	85	93	90	83	102	62	108	97	5	14	3	8	7	12	5	16	5	16	2	10	4	6	2	13	20
		10	75	107	116	89	69	62	90	68	88	103	95	91	101	70	92	88	6	6	4	9	4	5	3	11	4	11	3	7	4	2	3	14	18
7	PFDS	1	86	77	104	72	78	59	82	60	75	80	79	77	110	75	100	84	9	19	5	31	9	7	5	18	8	16	4	8	3	4	4	15	17
		10	60	94	106	84	53	48	88	54	75	98	87	71	103	63	80	70	6	9	4	10	3	7	5	14	7	13	5	7	3	5	10	11	24
8	PFDoDS	1	38	63	80	79	45	48	80	32	40	88	75	56	85	57	67	84	21	19	7	9	12	5	8	26	7	12	9	10	3	7	17	13	29
		10	45	78	71	73	42	41	84	43	47	90	73	53	95	53	45	89	5	13	9	5	4	18	6	14	6	12	7	5	4	4	31	14	31
9	PFBA	1	N.A.	111	96	94	95	90	94	102	125	83	110	119	104	80	91	112	N.A.	34	18	17	27	12	8	12	25	19	2	9	7	13	8	18	13
		10	88	172	104	110	100	103	95	100	99	94	106	116	107	92	104	103	8	3	3	11	4	3	4	1	3	12	2	8	1	3	5	4	18
10	PFPeA	1	112	105	90	92	101	74	93	102	91	112	103	127	104	100	98	108	3	9	14	22	9	10	2	5	15	2	4	5	3	8	2	9	12
		10	118	117	105	111	96	90	93	100	97	104	97	96	105	98	104	103	3	2	4	10	3	4	5	1	2	10	1	4	2	2	1	4	8
11	PFHxA	1	107	99	99	95	96	87	97	105	91	113	94	109	100	96	101	121	14	9	4	17	8	6	6	10	15	5	6	6	6	14	2	2	9
		10	112	105	112	104	92	100	95	100	80	104	97	104	103	91	104	110	2	1	2	5	4	2	5	2	3	10	0	5	2	3	1	2	8
12	PFHpA	1	103	108	96	102	91	86	94	103	105	111	108	92	103	107	100	87	13	11	9	9	10	4	4	3	8	8	3	9	5	7	3	4	8
		10	117	118	108	104	92	91	93	100	108	103	103	85	102	98	103	102	2	11	5	6	4	2	5	1	8	10	2	8	2	3	1	2	9
13	PFOA	1	101	108	108	102	103	81	74	101	69	115	99	106	104	99	102	37	16	24	9	18	16	15	13	6	6	8	14	14	6	10	3	29	21
		10	114	112	110	104	117	94	89	100	71	108	89	90	106	100	103	108	5	16	3	4	5	1	5	3	1	10	5	5	4	3	1	1	12
14	PFNA	1	120	123	99	85	97	74	95	96	97	129	99	94	111	87	99	86	9	10	5	16	9	11	7	5	10	9	4	5	4	10	2	20	15
		10	118	124	111	100	97	90	93	101	98	115	105	91	110	96	103	121	4	9	4	4	3	3	6	1	6	10	4	7	3	5	1	3	10
15	PFDA	1	122	96	96	102	91	81	101	104	86	127	101	104	102	98	98	96	7	6	6	9	3	11	5	8	11	11	1	4	9	7	1	86	11
		10	118	120	105	109	89	77	93	99	86	123	102	96	105	95	101	136	6	10	5	6	5	10	6	3	3	11	1	5	3	6	1	4	15
16	PFUnDA	1	103	122	83	107	89	82	92	115	83	127	100	109	85	104	116	97	10	2	10	11	4	10	13	17	4	13	1	4	3	7	2	7	14
		10	118	139	107	105	86	76	93	105	84	127	96	114	104	98	100	114	2	5	5	4	4	5	3	8	4	8	3	5	1	5	1	4	16
17	PFDoDA	1	112	90	88	95	97	109	102	113	88	129	96	112	100	96	112	93	24	9	3	11	3	6	3	16	3	13	2	2	3	5	1	32	11
		10	118	123	106	104	93	77	94	106	94	136	93	86	108	95	100	105	4	5	3	6	3	5	5	12	3	9	1	4	6	1	1	19	14
18	PFTrDA	1	107	115	78	93	85	94	107	97	90	92	106	105	81	108	94	89	15	4	14	17	2	6	2	17	8	16	3	5	6	5	8	12	11
		10	100	126	100	97	82	71	97	95	92	119	103	82	102	98	71	104	5	11	4	4	4	16	9	12	5	13	3	2	3	4	13	11	15
19	PFTeDA	1	117	106	83	90	95	78	102	114	90	129	90	85	88	88	108	99	27	9	6	3	5	9	1	26	8	12	3	15	4	10	2	18	14
		10	115	127	109	107	91	88	94	117	121	139	95	76	103	99	98	94	5	4	5	7	3	5	4	18	6	13	3	9	5	4	2	19	16
20	PFHxDA	1	158	100	23	118	95	40	102	85	117	110	44	55	87	119	69	84	20	8	14	15	5	6	3	46	5	11	29	9	4	3	35	14	40
		10	121	149	37	123	95	64	94	107	163	138	43	60	102	119	91	102	10	2	43	6	3	20	3	29	4	9	39	7	6	4	43	8	36
21	PFOcDA	1	173	122	12	126	111	56	100	86	121	128	20	49	92	115	69	73	21	12	8	8	2	8	3	52	7	11	87	12	10	3	35	8	47
		10	156	173	1	124	116	79	98	119	145	138	8	42	97	138	91	91	7	2	34	3	5	18	3	38	5	18	99	3	7	11	43	11	49

N.A.: Not available because of the negative value obtained by subtracting the concentration of the blank tap water sample from that of PFAA-added tap water sample.

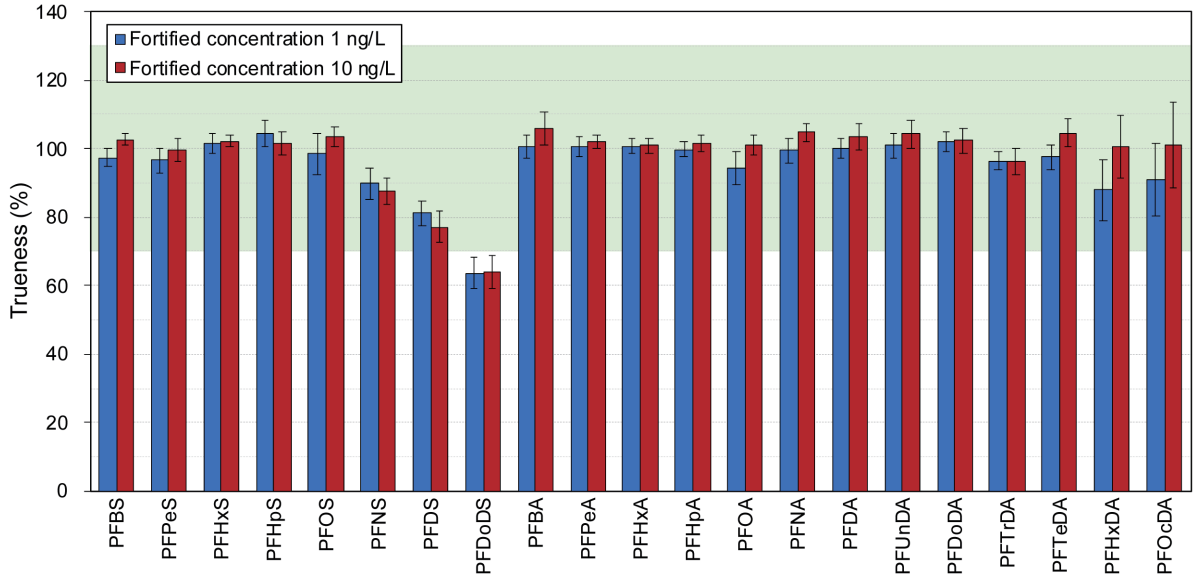


Fig. 5 Truness of each PFAA analytes in all laboratories corrected by ^{13}C-PFAA-ES (Average ± S.E.).

the intermediate precision satisfies the target value when the reproducibility is less than the target value associated with intermediate precision [25]. In this study, RSD_R was evaluated on the basis of the concepts described above. The RSD_R of each PFAA corrected by ^{13}C-PFAA-ES satisfies the target value of intermediate precision except for PFDoDS, PFHxA, and PFOcDA (**Table 7**). Conversely, the RSD_R of each PFAA without recovery correction by ^{13}C-PFAA-ES does not satisfy the target value of intermediate precision in many laboratories (**Table 5**). Therefore, recovery adjustment by use of a surrogate with similar recovery behavior to the PFAA analyte is required to achieve satisfactory PFAA quantification.

CONCLUSIONS

A LC/MS/MS analytical method for the simultaneous determination of 21 PFAAs has been developed to facilitate future drinking water regulations in Japan. An inter-laboratory test comprising 16 laboratories using different instrument was conducted to establish the applicability of the method to drinking water analysis from various sources in Japan. Calibration curves over 1–50 µg/L or 1–20 µg/L concentration ranges were constructed for all PFAAs. All analytes were detectable at 1 µg/L level in all laboratories. Several laboratories detected some PFAAs at concentrations up to 13 ng/L in the blank tap water samples. However, PFAA concentrations of the operation blank samples were generally below the 1 ng/L LOQ. Correction for the presence of PFAAs in the blank tap water samples was accomplished by the subtraction of the PFAA peak area of the blank sample from that of the PFAA-fortified tap water sample prior to the calculation of the concentrations. The trueness, RSD_r, and RSD_R of the PFAA analyte were satisfactory in most cases when the recoveries were corrected by the ^{13}C-PFAA-ES. Thus, satisfactory PFAA quantification requires recovery adjustment using surrogates with similar recovery characteristics to the PFAA analytes.

ACKNOWLEDGEMENTS

This work was supported by the Health and Labour Sciences Research Grant (Grants no. 19LA1005 and 22LA1007) of the Ministry of Health, Labour, and Welfare, Japan.

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